

Earth's Gravity Redirects Juice Toward Jupiter

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A planet can help a spacecraft change its speed and direction without requiring a large fuel burn. On September 28, the European Space Agency's Jupiter Icy Moons Explorer, known as Juice, used Earth's gravity to reshape its route toward Jupiter.

The **maneuver**, called a gravity assist, changed Juice's direction by 20 degrees relative to its previous path and increased its speed by 3.5 kilometers per second. Getting that result required a precisely planned approach. During the four weeks before the flyby, controllers used only one of six available opportunities to make a small course correction.

Juice reached its closest point at 11:45 UTC, passing 8,640 kilometers above the Indian Ocean. Its monitoring cameras captured a sequence of images during the encounter. Mission controllers began particularly close tracking on August 17 and planned to continue through October 10.

Spacecraft Operations Manager Angela Dietz said the careful **navigation** used only a small amount of the fuel reserved for the flyby. The saved fuel can support later observations of Jupiter's icy moons.

Changing course was the main purpose of the encounter, but Earth also provided a familiar planetary target for testing scientific equipment. Teams spent months coordinating activities for Juice's 10 instruments. **Calibration** means checking and adjusting an instrument so that its measurements can be interpreted reliably.

These opportunities are limited during the long journey. Earlier chances included the 2024 lunar-Earth gravity assist and observations of Comet 3I/ATLAS in 2025. For this encounter, scientists **prioritized** checks that would best prepare the instruments for Jupiter, while considering whether particular activities could happen later.

Power also shaped the schedule. When Juice entered Earth's shadow, it relied on its batteries alone, affecting which activities could take place. The visit gave researchers a chance to learn how the instruments behave in space and refine tools for analyzing their data.

Juice also spent several days traveling through Earth's **magnetotail**, the long extension of our planet's magnetic field pointing away from the Sun.

The spacecraft is scheduled to make 35 **flybys** of Ganymede, Callisto and Europa. Those encounters will demand more complicated scientific planning than the Earth visit. Discussions about instrument operations for Europa began about 10 years ago and will continue until the planned passes in the early 2030s. This Earth encounter therefore served as both a fuel-saving **maneuver** and an important rehearsal.

Word Treasure

maneuver -- A planned movement that changes a vehicle's path or position.

navigation -- Planning and controlling the route a vehicle follows.

Calibration -- Checking and adjusting an instrument to make its measurements reliable.

prioritized -- Chose certain activities to come before others.

magnetotail -- The long part of Earth's magnetic field extending away from the Sun.

flybys -- Passes near a planet, moon or other object in space.

Background you need

Calibration helps researchers interpret measurements from scientific instruments reliably.
A flyby can change a spacecraft's route and provide an opportunity to test instruments.
Earth's magnetotail is an extension of its magnetic field pointing away from the Sun.

Break it down

Introduces Juice and the purpose of its Earth gravity assist.
Reports changes in speed and direction and the navigation required.
Describes the closest approach, images and tracking schedule.
Attributes the explanation of fuel savings to Angela Dietz.
Explains calibration opportunities, priorities and power constraints.
Describes the magnetotail passage and future moon flybys.

Why it matters

The flyby conserved fuel for observations at Jupiter and provided a rehearsal for instrument operations during the planned encounters with its icy moons.

Test what you remember

1. What was the main purpose of Juice's Earth encounter?
 - (A) To begin its planned moon flybys
 - (B) To change its course toward Jupiter
 - (C) To observe the comet 3I/ATLAS
 - (D) To study the surface of Europa
2. How much did the gravity assist increase Juice's speed?
 - (A) 3.5 kilometers per second
 - (B) 10 kilometers per second
 - (C) 20 kilometers per second
 - (D) 6 kilometers per second
3. How many course corrections did controllers make during the four weeks before the flyby?
 - (A) Four small corrections
 - (B) Six small corrections
 - (C) Two small corrections
 - (D) One small correction
4. What did Angela Dietz say about fuel use during the flyby?
 - (A) Taking camera images required extra reserved fuel.
 - (B) Instrument testing used all of the reserved fuel.
 - (C) Careful navigation used little of the reserved fuel.
 - (D) Battery operation replaced the need for reserved fuel.
5. Which instrument checks did scientists prioritize for this encounter?
 - (A) Checks that prepared the instruments for Venus
 - (B) Checks that prepared the instruments for Mars
 - (C) Checks that prepared the instruments for Jupiter
 - (D) Checks that prepared the instruments for Saturn
6. What is Earth's magnetotail?
 - (A) A magnetic field extension pointing away from the Sun
 - (B) A spacecraft route connecting Earth with its Moon
 - (C) An atmospheric layer surrounding the Indian Ocean
 - (D) A stream of camera images sent back to Earth

Answer key (try the quiz first!): 1: B 2: A 3: D 4: C 5: C 6: A

1. Angela Dietz: She said careful planning and precise navigation kept fuel use low during the flyby, leaving more fuel for observations of Jupiter's icy moons.